Dev Sanjay Sakaria

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Summary

Robotics and Product Development Engineer with hands-on experience designing and validating electromechanical systems for surgical and medical applications. Strong background in force-controlled robotics, system architecture, and human–robot interaction, including teleoperation and haptic interfaces. Experienced in ROS-based system integration, Robot Kinematics and Haptic Technology with a focus on safety, reliability, and manufacturability for Class III medical devices.

Education

University of Illinois, Urbana-Champaign, Grainger College of Engineering

M.Eng. in Autonomy & Robotics | Aug 2024 – Dec 2025 | GPA: 3.71/4.00

- **Relevant Coursework:** Autonomous Vehicles Systems Engineering, Principles of Safe Autonomy, Robotics, Applied Machine Learning, Soft Robotics, Surgical Technologies.

Manipal Institute of Technology

B.Tech in Mechatronics Engineering | Jun 2020 – Jul 2024 | GPA: 3.42/4.00

- **Relevant Coursework:** Mechanical Design, Controls, DSP, AI, Hydraulics , Pneumatics

Experience

Healthcare Technology Innovation Center, IITM Research Park – Robotics *Engineer*

Chennai, India · Jan 2026 – Sep 2026

- Developing an admittance-based force control framework for a 7-DOF robotic manipulator, enabling compliant interaction and stable probe–tissue contact for robot-assisted ultrasonography.
- Designing and implementing a real-time external control layer to augment the robot's internal controller, improving force tracking, responsiveness, and task-specific performance.
- Architecting and prototyping a grounded haptic master interface with impedance-based feedback for intuitive and safe teleoperation of the robotic system.
- Modeling system dynamics and tuning control parameters to ensure stability, safety, and reliable human–robot interaction in a clinical environment.

Hourglass Medical – Robotics *Systems Design Intern*

Champaign, IL, USA · May 2025 – August 2025

- Led system architecture design of a multi-DOF robotic surgical lighting platform using MBSE (SysML), defining kinematics, perception, control, and actuation interfaces.
- Implemented ROS2-based perception and planning nodes, integrating real-time communication via CAN bus with sensors and actuators.
- Modeled and optimized multi-DOF robotic kinematics to improve positioning accuracy under real-world constraints.

Robert Bosch Center for Cyber-Physical Systems – *Robotics Intern*

Bengaluru, India · Jan 2024 – Jun 2024

- Developed multi-robot localization pipelines using ROS, EKF-based sensor fusion, and vision-based estimation.
- Integrated perception modules (YOLO, OpenCV, PnP) for robust 6D pose estimation.
- Improved localization accuracy by **39%** through systematic tuning and validation.
- Validated system performance using simulation (Gazebo, IsaacSim) and documented results for reproducibility.

Sansera Engineering - Automation Intern

Bengaluru, India · Dec 2022 – Feb 2023

- Designed and modeled an automated bearing assembly system, supporting fabrication and assembly. .
- Developed PLC control logic and HMI interfaces to improve operator efficiency and reduce cycle time.
- Integrated and debugged industrial vision systems for reliable inline inspection.

Skills

Haptics, ROS/ROS2, Kinematics, Force/Impedance Control, Teleoperation | Python, C++, MATLAB | Gazebo, Isaac Sim |SolidWorks | Sensors/Actuators, CAN, PLC | CV (OpenCV, YOLO), EKF

Certification

ISO 14971:2019 – Medical Device Risk Management (Udemy, 2024)

Project: Simulink/Simscape servo model + HIL test bench .